

Mechanical Selection and Justification

Encapsulator Design Sources and Attribution

The AgriGuard Encapsulator was developed using existing open-source 3D-printable enclosure designs as reference designs rather than directly copying an existing enclosure. These designs were studied to understand common approaches to enclosure construction, including removable covers, internal mounting points, cable openings, ventilation and screw fastening.

Reference designs were obtained from publicly available open-source repositories and 3D-printing communities, including:

- **Printables – ESP32 DevKit V1 Case:** An open-source ESP32 enclosure used as a reference for protecting and mounting an ESP32 development board. The design is licensed under GNU GPL v3.0.
- **Printables – Roomy ESP32 Enclosures:** A larger ESP32 enclosure concept that provides additional internal space for wiring and electronics. It uses a CC BY-NC-SA 4.0 license.
- **Printables – Capacitive soil moisture Encapsulator:** A two-piece enclosure consisting of a printed base and removable cover, providing another reference for the physical arrangement of an ESP32 enclosure. It uses a CC BY-NC-SA license.

These designs served as engineering references and sources of design inspiration. The AgriGuard Encapsulator was subsequently adapted and redesigned around the project's specific hardware, including the ESP32 controller, relay/MOSFET circuitry, sensor connections, emergency-stop system, power connections and required cable-routing locations.

The final enclosure therefore represents a project-specific design rather than an unmodified reproduction of an existing model. Dimensions, mounting positions, openings and component arrangement were selected according to the requirements of the Agriguard system.

All external designs used for reference are acknowledged to their respective creators, and their individual licences are respected when using, modifying or redistributing any of their original CAD files.

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Component Fastening

Components are mounted using integrated mounting bosses/standoffs that are incorporated into the 3D-printed enclosure. The ESP32 development board is mounted onto raised standoffs using small machine screws.

The relay/MOSFET control circuitry is mounted using separate screw holes or standoffs.

The LCD/display, where applicable, is secured into a dedicated opening in the enclosure.

Larger components are positioned so that they do not interfere with one another or with the enclosure cover.

Screws allow components to be removed for maintenance or replacement without destroying the enclosure.

Electrical Connections

The electronics are connected using insulated wires and removable connectors, rather than permanently soldering every component directly to the enclosure.

Cable-entry point is incorporated into the enclosure for connections to:

- ✓ Soil-moisture sensors
- ✓ Water-flow sensor
- ✓ Water-level sensor
- ✓ Irrigation valves/pump
- ✓ Power supply
- ✓ Emergency-stop circuit

The cable opening can be fitted with rubber grommets or cable glands to reduce cable movement and provide additional protection against dust and moisture.

Enclosure Assembly

The general assembly process is:

CAD design → 3D print → install threaded inserts → mount electronics → connect wiring → close enclosure → test

The enclosure is designed so that the electronics can be assembled, inspected and serviced without having to redesign or reprint the entire housing.